Signals Project Report

Tutorial Number: 14

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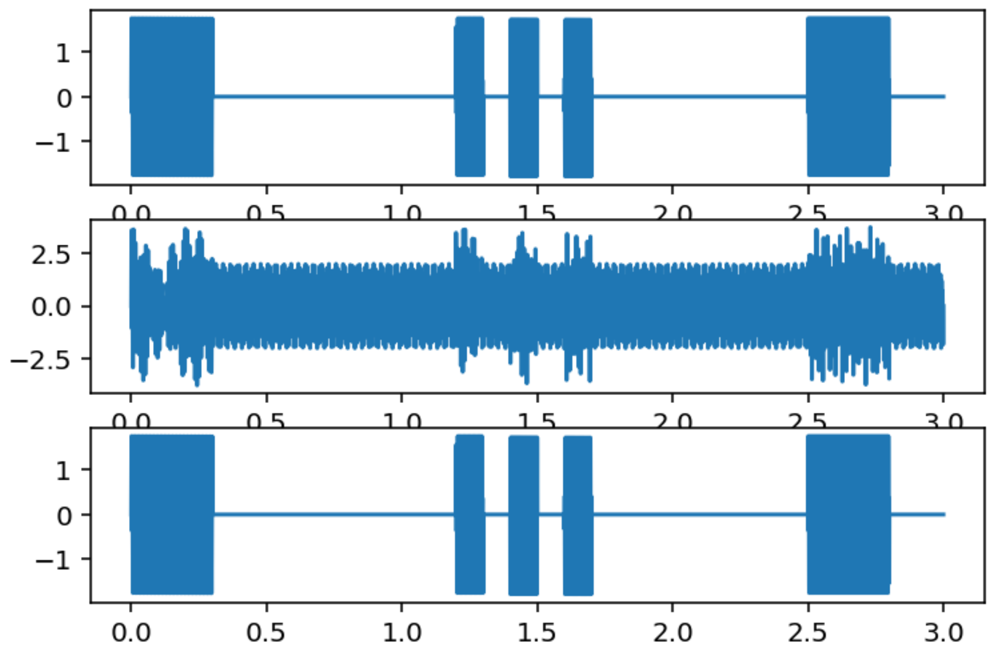
Code:

```
import numpy as np
import matplotlib.pyplot as plt
import sounddevice as sd
from scipy.fftpack import fft

t = np.linspace(0,3,3*44100)

F_array = np.array([130.81,146.83,164.81,174.61, 196.00, 220.00, 246.94])
f_array= np.array([261.63, 293.66, 329.63, 349.23, 392.00, 440.00, 493.88])

t_array = np.array([0, 1.2, 1.4, 1.6, 2.5, 3.7, 3.9, 4.1])
T_array = np.array([0.3, 0.1, 0.1, 0.1, 0.3, 0.1, 0.1, 0.1])

N=5
counter=0
x=0
while (counter<N):
        Fi = F_array[counter]
        fi = f_array[counter]
        ti = t_array[counter]
        Ti = T_array[counter]
        x = x + (np.sin(2*np.pi*Fi*t)+np.sin(2*np.pi*fi*t))*((t>=ti)&(t<=(ti+Ti)))
        counter = counter+1

#plt.plot(t, x)
sd.play(x,3*44100)

#ADD NOISE
N = 3*44100
f = np. linspace(0 , 44100/2 , int(N/2))

x_f = fft(x)
x_f = 2/N * np.abs(x_f[0:int(N/2)])

fn1 , fn2 = np.random.randint(0, 512, 2)
n = np.sin(2*np.pi*fn1*t)+np.sin(2*np.pi*fn2*t)

xn = x+n
xn_f = 2/N * np.abs(fft(xn)[:N//2])
```

```
# REMOVE NOISE
(z,) = np.where(xn_f>int(np.ceil(np.max(x_f))))
cancel_noise =0
for i in z :
    z=f[i]
    cancel_noise+=np.sin(2* np.pi * np.round(z) * t)
    x_filtered = xn - cancel_noise

xFiltered_f = fft(x_filtered)
xFiltered_f = 2/N * np.abs(xFiltered_f [0:int(N/2)])

sd.play(x_filtered, 3 * 44100)

#TIME DOMAIN
plt.figure()
plt.subplot(3,1,1)
plt.plot(t,x)
plt.subplot(3,1,2)
plt.plot(t,xn)
plt.subplot(3,1,3)
 plt.plot(t,x_filtered)

#FREQUENCY DOMAIN
plt.figure()
plt.subplot(3,1,1)
plt.plot(f,x_f)
plt.subplot(3,1,2)
plt.plot(f,xn_f)
plt.subplot(3,1,3)
plt.plot(f,xFiltered_f)
```

Graphs:

Time Domain Plots



Frequency Domain Plots